A

Mini Project Report

on

Health Monitoring

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science and Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **Health Monitoring** has been submitted by Akash Yadav (24107037), Aman Yadav (24107051) Anushka Kadam (24107087) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science and Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Prof Poonam Pangarkar**. Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1
Introduction

Health monitoring plays a vital role in maintaining a healthy lifestyle and preventing serious medical conditions. In today's busy world, many people fail to regularly track important health parameters such as heart rate, blood pressure, blood sugar level, and daily physical activity. As a result, health issues often go unnoticed until they become severe. Traditional methods of maintaining health records manually are inconvenient, time-consuming, and prone to errors.

With the rapid growth of mobile technology and digital healthcare solutions, health monitoring can now be performed more efficiently and accurately through smart applications. Mobile applications provide users with the ability to monitor their health anytime and anywhere, store records digitally, and receive alerts in case of abnormal health conditions. These applications increase awareness and encourage users to take preventive actions at an early stage.

The Mobile-Based Health Monitoring System is designed to provide a simple, user-friendly, and effective solution for continuous health tracking. The application is developed using Flutter for cross-platform mobile development and Firebase as the backend database for secure cloud storage and real-time data management. Users can enter health-related data such as heart rate, blood pressure, and sugar levels, which are stored and analysed by the system.

One of the important features of the system is the ability to detect abnormal health conditions by comparing user-entered values with normal health ranges. Whenever abnormal values are detected, the application provides instant alerts and notifications so that users can take immediate action. The system also includes a medicine reminder feature that helps users follow their treatment schedule properly.

1.1 Purpose

The primary purpose of the Mobile-Based Health Monitoring System is to provide users with a simple, efficient, and reliable platform to monitor their daily health parameters regularly. In today's fast-paced lifestyle, people often neglect their health due to busy schedules, lack of

awareness, or difficulty in maintaining health records manually. This application helps overcome these problems by offering a digital solution for continuous health tracking.

The system is designed to help users monitor important health parameters such as heart rate, blood pressure, and blood sugar level through a mobile application. By maintaining these records digitally, users can access their health history anytime and anywhere without the need for manual paperwork.

Another important purpose of the system is to provide early detection of abnormal health conditions. The application analyzes the entered health values and compares them with normal ranges. If any abnormality is detected, the system generates alerts and notifications so that users can take preventive action at the right time.

The application also aims to improve medication management through a medicine reminder feature, which helps users follow their treatment schedule properly and avoid missing doses.

Furthermore, the system provides graphical representation of health data, making it easier for users to understand health trends and track improvements over time.

1.2 Problem Statement

In today's fast-moving lifestyle, many people do not regularly monitor important health parameters such as heart rate, blood pressure, blood sugar level, and daily physical activity**.** Due to busy schedules, lack of awareness, and the inconvenience of manual health tracking, health-related problems often remain unnoticed until they become serious.

Traditional methods of maintaining health records, such as writing reports manually or storing paper-based prescriptions, are inefficient and difficult to manage. These records can be misplaced, damaged, or forgotten, making it difficult for users to track their health history over time. In addition, manual tracking does not provide real-time analysis or immediate alerts when abnormal health conditions occur.

Many existing health applications provide limited features and do not offer a complete solution for continuous monitoring, digital record management, medicine reminders, and graphical health analysis in one platform. As a result, users may fail to take timely preventive action, which can lead to severe health complications.

Therefore, there is a need for a mobile-based health monitoring system that can continuously track health parameters, maintain digital records securely, provide alerts for abnormal conditions, send medicine reminders, and help users monitor their health efficiently anytime and anywhere.

1.3 Objectives

The main objectives of the Mobile-Based Health Monitoring System are as follows:

1. **To monitor important health parameters**
   The system aims to help users regularly track important health parameters such as heart rate, blood pressure, and blood sugar level through a mobile application for better health management.

2. **To detect abnormal health conditions and provide alerts**
   The system analyses the entered health values and detects abnormal conditions at an early stage. It provides instant alerts and notifications so that users can take timely preventive action.

3. **To maintain digital health records**
   The system stores health-related data securely in the Firebase cloud database, allowing users to access, manage, and review their health history anytime and anywhere.

4. **To improve healthcare management through reminders and graphs**
   The system provides medicine reminders and graphical representation of health data, helping users follow treatment schedules and understand their health trends over time.

1.4 Scope

The Mobile-Based Health Monitoring System is designed to provide a simple and efficient platform for users to monitor and manage their health regularly. The system mainly focuses on

personal healthcare monitoring and can also be expanded for professional medical use in the future. The scope of the system includes the following capabilities:

1. **Personal Health Monitoring**
   The system allows users to regularly monitor important health parameters such as heart rate, blood pressure, and blood sugar level through a mobile application. This helps users stay aware of their health condition and take preventive action whenever required.

2. **Digital Health Record Management**
   The application stores health-related data securely in the cloud database using Firebase. Users can access, manage, and review their previous health records anytime and anywhere, making it easier to maintain proper medical history.

3. **Alerts and Notifications**
   The system continuously analyzes the entered health data and detects abnormal conditions. If any health value exceeds the normal range, the application provides instant alerts and notifications so that users can take timely action.

4. **Medicine Reminder**
   The application includes a medicine reminder feature that helps users follow their treatment schedule properly by sending reminders to take medicines on time.

5. **Graphical Health Analysis**
   The system provides graphical representation of health data using charts and trends. This helps users understand their health progress and monitor improvements over time.

6. **Future Expansion**
   In the future, the system can be integrated with wearable devices for automatic real-time health monitoring. It can also be expanded for use in hospitals, clinics, and remote patient monitoring services.

Chapter 2
Proposed System

The proposed system, Mobile-Based Health Monitoring System, is designed to provide users with a smart, simple, and efficient way to monitor and manage their health regularly through a mobile application. In today's fast-paced lifestyle, many people fail to monitor their health parameters regularly due to busy schedules or lack of awareness. This system helps users continuously track important health parameters such as heart rate, blood pressure, and blood sugar level in real time. It provides an easy and convenient digital solution for maintaining personal health records and improving overall health awareness.

The application is developed using Flutter, which provides a modern, responsive, and user-friendly interface for mobile devices. Firebase is used as the backend database to securely store user details and health records in the cloud. Users can manually enter their health data through the application, and the data is stored safely for future reference.

The proposed system also maintains digital health records and provides graphical representation of health data through charts and trends. This helps users understand their health progress and monitor improvements over time. In addition, the application includes a medicine reminder feature that helps users follow their treatment schedule properly by sending timely reminders

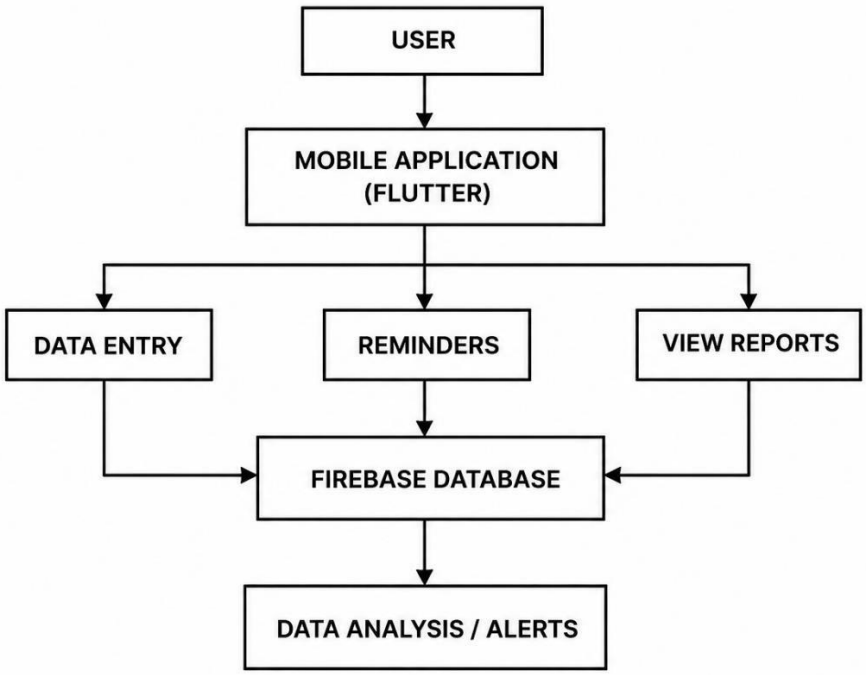


Figure 2.1 Block Diagram

2.1 Features and Functionality

The Mobile-Based Health Monitoring System provides various features and functionalities that help users monitor and manage their health effectively. The major features of the system are as follows:

1. **User Login and Authentication**
   The system provides a secure login and authentication feature that allows users to safely access their personal health information. It protects user privacy and ensures that only authorized users can view or update health records

2. **Health Data Entry**
   Users can enter important health parameters such as heart rate, blood pressure, and blood sugar level through the mobile application. The entered data is stored securely and used for continuous monitoring and analysis.

3. **Digital Health Records**
   The system maintains digital records of the user's health history for future reference. Users can access and review their previous health data anytime and anywhere through the application.

4. **Alerts and Notifications**
   The application continuously checks the entered health values and compares them with normal ranges. If any abnormal condition is detected, the system sends alerts and notifications to the user for immediate action.

5. **Medicine Reminder**
   The system provides medicine reminders to help users follow their treatment schedule properly. Timely notifications ensure that users do not miss important doses of medicine.

Chapter 3
Project Outcomes

The Mobile-Based Health Monitoring System successfully provides an efficient, reliable, and user-friendly solution for monitoring important health parameters through a mobile application. The project helps users track health data such as heart rate, blood pressure, and blood sugar level in a simple and convenient way. It improves health awareness by encouraging users to regularly monitor their health and take preventive actions whenever required. The easy-to-use interface makes the application suitable for users of different age groups.

The system stores health-related data digitally using Firebase cloud storage, allowing users to access, manage, and review their health records anytime and anywhere. This removes the need for manual paperwork and ensures secure and efficient data management. Users can maintain proper medical history and track their previous records for future reference.

The application helps in the early detection of abnormal health conditions by continuously analysing the entered health values and comparing them with normal health ranges. If any abnormal value is detected, the system immediately generates alerts and notifications. This enables users to take timely action and reduces the chances of serious health complications.

The medicine reminder feature ensures that users follow their treatment schedule properly by sending timely reminders and notifications. This is especially useful for patients and elderly users who need regular medication. The reminder system helps users maintain discipline in their healthcare routine. In addition, the application provides graphical representation of health data through charts and reports. This helps users understand their health trends, monitor improvements, and analyse changes over time. Overall, the project demonstrates how mobile technology and cloud-based services can be effectively used for continuous health monitoring and better personal healthcare management. It promotes a healthy lifestyle and provides a practical digital healthcare solution.

Chapter 4
Software Requirements

This chapter describes the software, tools, and technologies required for the development and deployment of the Mobile-Based Health Monitoring System. These technologies help in building a secure, efficient, and user-friendly application for continuous health monitoring.

1. **Operating System**
   • Android OS – Used for running the mobile application on Android devices.
   • Windows/macOS – Used as the development environment for coding, testing, and deployment.

2. **Development Tools**
   • Flutter SDK / Android Studio / Visual Studio Code – Used for designing, coding, and testing the application.
   • Firebase Console – Used for backend services and cloud database management.

3. **Programming Language**
   • Dart – Used as the primary programming language for developing the application in Flutter.

4. **Libraries and Packages Used**
   • Firebase Core – For connecting the app with Firebase services.
   • Firebase Auth – For user login and authentication.
   • Cloud Fire store – For storing health records in the cloud.
   • Flutter Local Notifications – For reminders and alerts.
   • FL Chart – For charts and graphical representation.
   • Shared Preferences – For local storage of user settings.

These software requirements ensure smooth performance, secure data storage, and a better user experience.

Chapter 5
Project Design

This chapter describes the overall system architecture, data flow, and component interactions of the Mobile-Based Health Monitoring System. It explains how different modules work together to provide continuous health monitoring and better healthcare management.

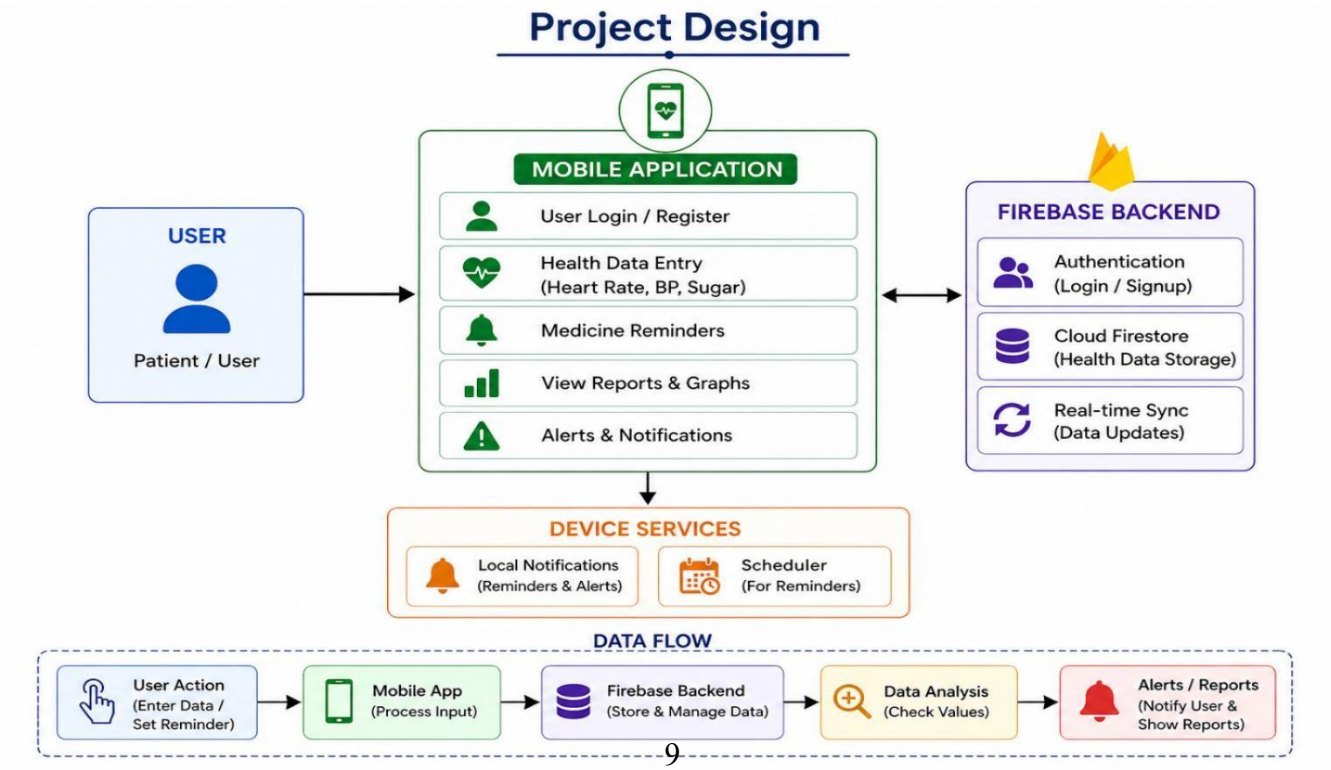
The system is designed with three main components: the mobile application, the backend database, and device-level services. The mobile application is developed using Flutter and provides a user-friendly interface for entering health data, viewing reports, and receiving alerts.

The system includes different modules such as User Authentication Module, Health Monitoring Module, Reminder Module, Alert Module, and Data Visualization Module. These modules work together to monitor health parameters, detect abnormal conditions, and provide useful insights to users.

Data Flow

User Action → Mobile App → Firebase Backend → Data Analysis → Alerts / Reports

This design ensures secure data storage, real-time monitoring, and smooth communication between all system components.



Chapter 6
Project Scheduling

Project scheduling defines the timeline, task allocation, and responsibilities of each team member throughout the development lifecycle of the Mobile-Based Health Monitoring System. It helps in proper planning, execution, and completion of the project within the given time period.

In the initial phase, the team focused on project inception, which included topic selection, requirement analysis, and discussion of system functionalities. During this stage, the project scope and objectives were clearly defined.

Chapter 7
Results

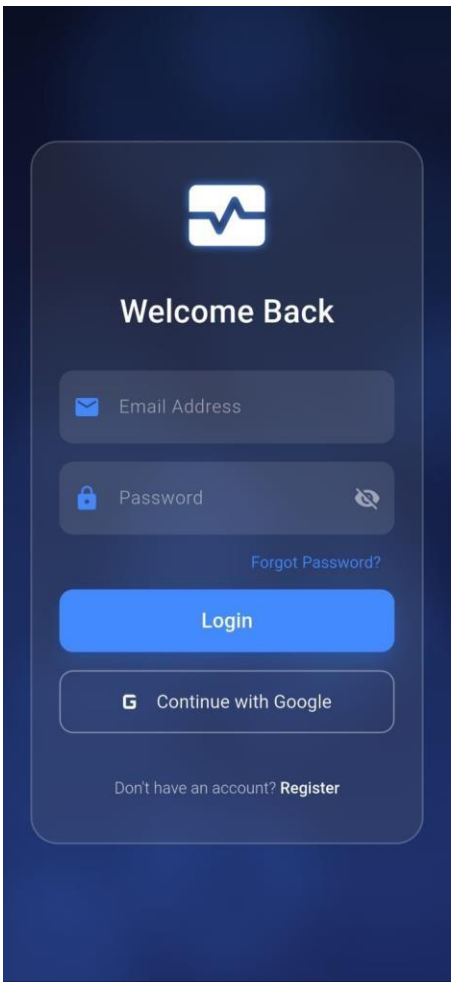


Fig 7.1 Login Page

In fig 7.1, the Login Page acts as the entry point of the Mobile-Based Health Monitoring System. It provides a clean, modern, and user-friendly interface for users to securely access the application. The page includes input fields for email address and password, allowing registered users to log in easily.

The page also includes a “Forgot Password?” option, which helps users recover their account in case they forget their password. A “Continue with Google” option is provided for quick and convenient login using Google authentication.

Additionally, users who do not have an account can register through the “Register” option available at the bottom of the page. The attractive and simple design improves user experience and makes navigation easier

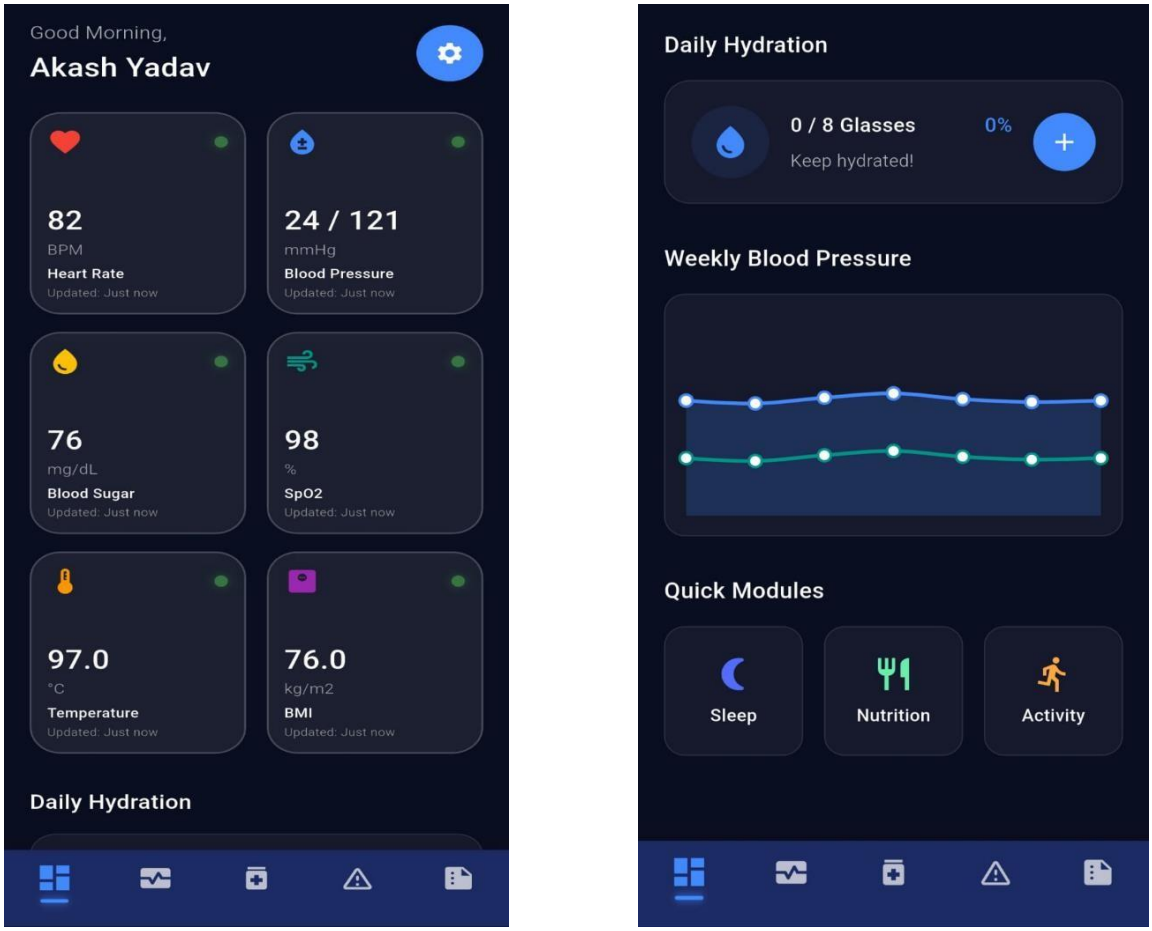


Fig 7.2 Dashboard Page

In fig 7.2, the Dashboard Page provides users with a complete overview of their health status time in a single interface. It displays important health parameters such as heart rate, blood pressure, blood sugar, SpO2 level, body temperature, and BMI in separate cards for easy monitoring. Each card shows the latest updated values, helping users track their health in real.

The dashboard also includes a Daily Hydration section, which helps users track their water intake and maintain proper hydration throughout the day. A Weekly Blood Pressure graph is provided to show trends and variations in blood pressure over time, allowing users to analyse their health progress.

In addition, the Quick Modules section provides shortcuts for features such as sleep monitoring, nutrition tracking, and activity monitoring. The bottom navigation bar allows users to quickly switch between different sections of the application. The dashboard is designed to be modern, user-friendly, and interactive for better healthcare management

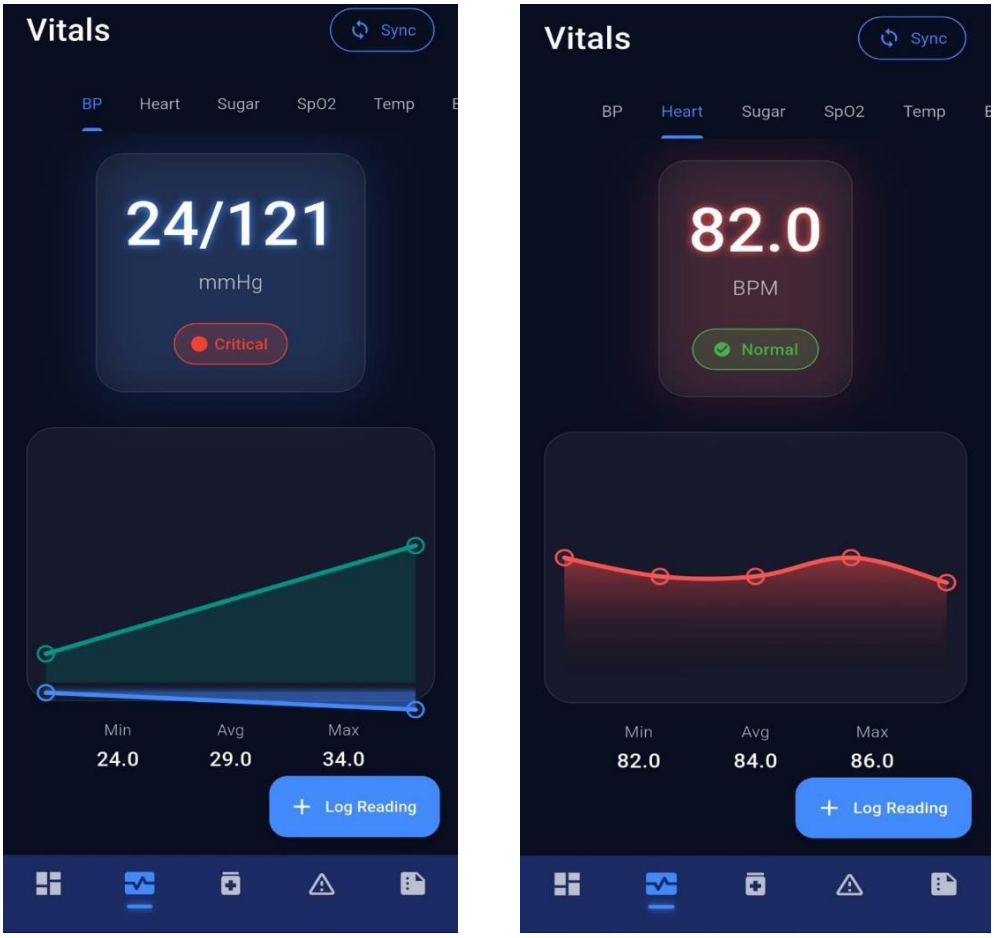


Fig 7.3 Vitals Page

In fig 7.3, the Vitals Page allows users to monitor detailed health parameters individually. The page contains different tabs such as Blood Pressure, Heart Rate, Sugar, SpO2, and Temperature, enabling users to switch between different health metrics easily.

Each section displays the current reading in a highlighted card along with the unit of measurement and health status such as “Normal” or “Critical.” This helps users quickly understand their current health condition. The system also provides graphical representation of previous readings in the form of charts, making it easier to analyse trends over time.

The page shows minimum, average, and maximum values for better analysis and comparison of health records. In addition, the “Log Reading” button allows users to manually add new readings for continuous monitoring.

The Vitals Page provides a detailed and interactive way to track individual health parameters and helps users take timely action if abnormal conditions are detected.

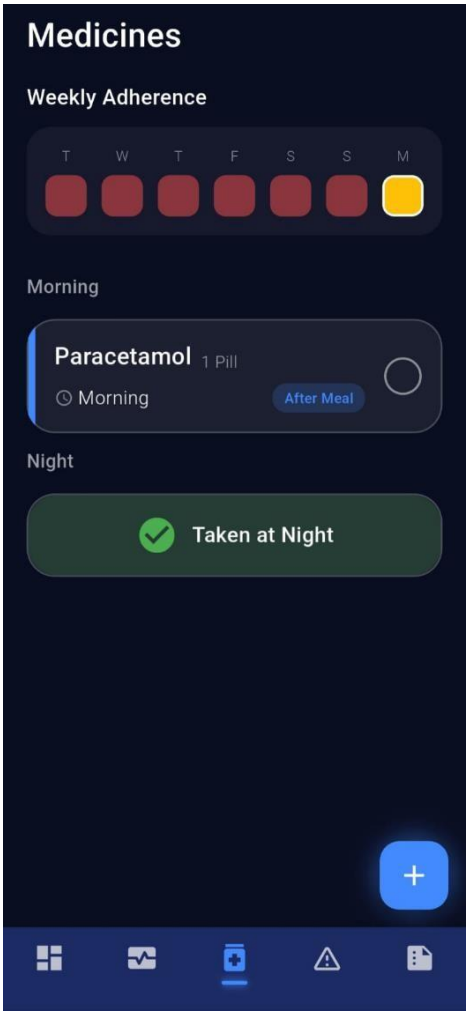


Fig 7.4 Medicine Reminder Page

In fig 7.4, the Medicines Page helps users manage their medication schedule effectively. It displays weekly adherence, allowing users to track whether medicines were taken on specific days of the week. This helps users maintain consistency in following their prescribed treatment.

The page shows medicine details such as the medicine name, dosage, and the time at which it should be taken, such as morning or night. It also provides additional instructions like “After Meal” to ensure proper medication intake.

A status indicator is provided to mark whether the medicine has been taken or not. In the given screenshot, the system shows “Taken at Night,” confirming successful completion of the scheduled dose.

Additionally, the page includes an add button that allows users to add new medicines and reminders easily. This feature helps users maintain discipline in their medication routine and improves

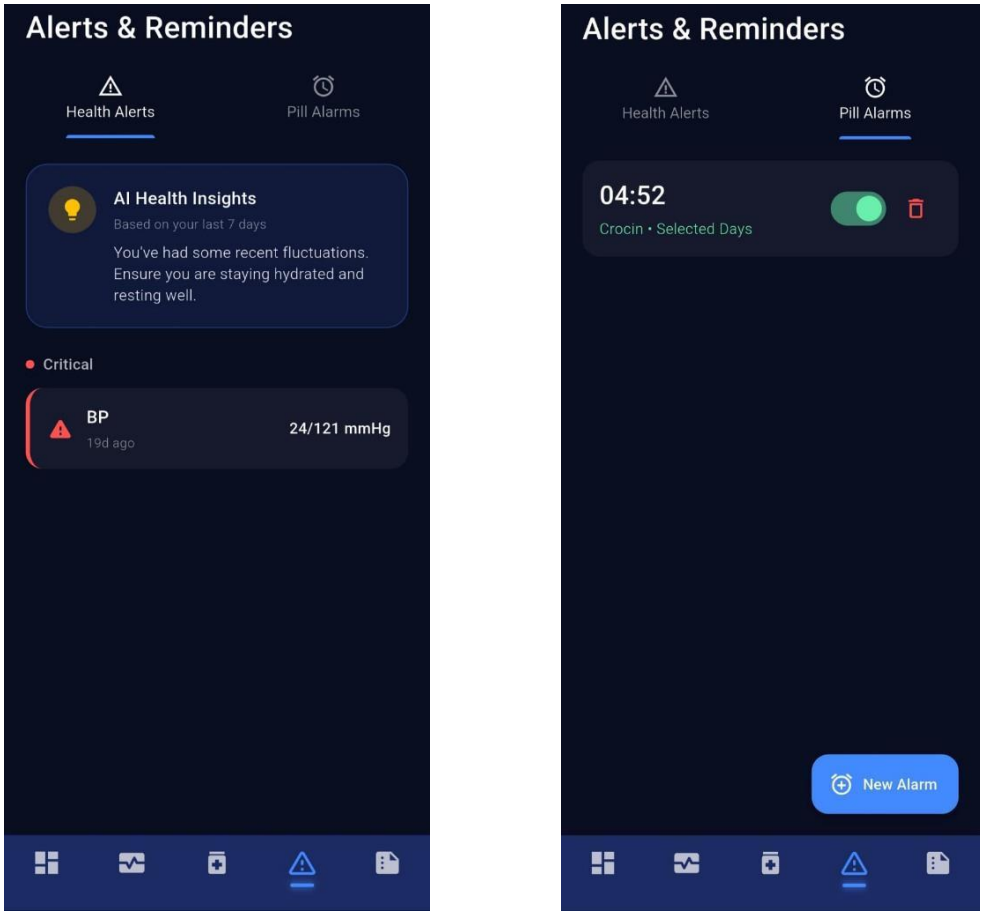


Fig 7.5 Alerts and Reminders Page

In fig 7.5, the Alerts and Reminders Page help users stay informed about critical health conditions and medication schedules. The page is divided into two sections: Health Alerts and Pill Alarms, allowing users to switch between health-related notifications and medicine reminders easily.

The Health Alerts section displays AI-generated health insights based on the user's recent health data. It provides useful suggestions and warnings if unusual fluctuations or abnormal conditions are detected. Critical alerts, such as abnormal blood pressure readings, are highlighted to grab the user's attention and encourage immediate action.

The Pill Alarms section allows users to set and manage medicine alarms for specific times and selected days. Users can enable or disable alarms using the toggle switch and delete alarms when they are no longer needed. A "New Alarm" button is also provided to create additional reminders.

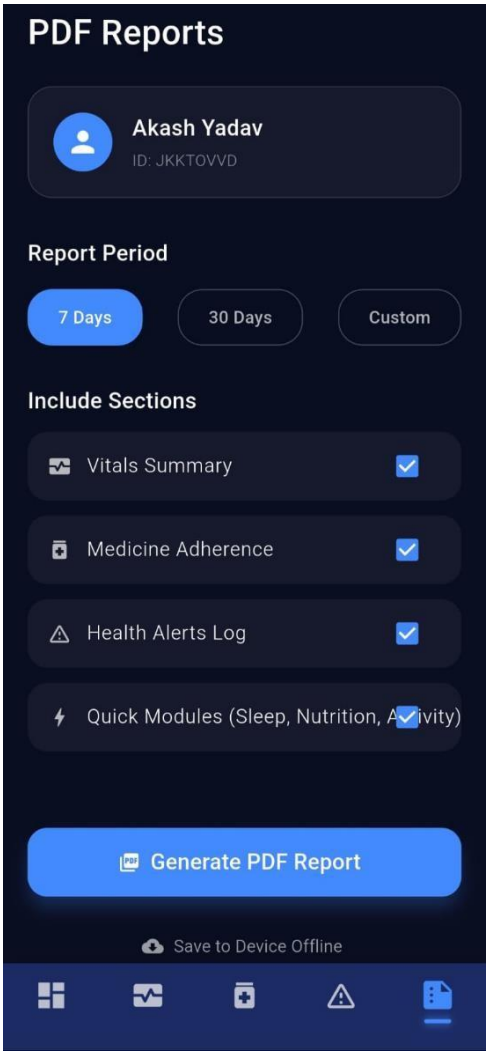


Fig 7.6 PDF Report Page

In fig 7.6, the PDF Reports Page allows users to generate and download health reports in PDF format for a selected time period. This feature helps users maintain digital records and share reports with doctors or healthcare professionals when required.

The page provides options to select the report duration, such as 7 days, 30 days, or a custom period. Users can choose the specific sections they want to include in the report, such as vitals summary, medicine adherence, health alerts log, and quick modules like sleep, nutrition, and activity.

After selecting the required options, users can generate the PDF report by clicking the “Generate PDF Report” button. The generated report can also be saved offline on the device for future reference.

This feature makes health record management easier, organized, and more convenient for users.

Chapter 8
Conclusion

The Mobile-Based Health Monitoring System is a smart, efficient, and user-friendly healthcare application designed to help users monitor and manage their health regularly through a mobile device. In today's busy lifestyle, people often ignore their health due to lack of time and awareness. This system provides a simple and convenient platform for tracking important health parameters such as heart rate, blood pressure, blood sugar level, SpO2, body temperature, BMI, and hydration levels in real time. It helps users stay aware of their health condition and encourages them to maintain a healthy and disciplined lifestyle.

The application includes various useful features such as digital health record management, medicine reminders, health alerts, hydration tracking, graphical analysis, and PDF report generation. These features make healthcare management easier and more organized. Users can maintain their complete medical history digitally, analyse health trends through graphs, and generate reports that can be shared with doctors or healthcare professionals when required. The medicine reminder and pill alarm features help users follow their treatment schedule properly and reduce the chances of missing important doses.

The system also provides alerts and notifications whenever abnormal health conditions are detected. AI-based health insights help users understand recent fluctuations in their health data and take preventive actions on time. This early detection and alert mechanism can reduce health risks and improve overall healthcare management.

The use of Flutter for frontend development and Firebase for backend services ensures that the application is secure, responsive, and reliable. Firebase Authentication provides secure login and registration, while Cloud Firestore stores health-related data safely in the cloud. The attractive user interface and simple navigation make the application easy to use for people of different age groups. Overall, this project successfully demonstrates how mobile technology and cloud-based services can be used for continuous health monitoring and better personal healthcare management. It reduces manual effort, improves health awareness, and provides a practical digital healthcare solution for modern users. In the future, the system can be enhanced by integrating wearable devices, real-time sensors, and advanced AI-based prediction systems for more accurate and automatic health monitoring.

Reference

1. Flutter Documentation – https://flutter.dev/docs
2. Firebase Documentation – https://firebase.google.com/docs
3. Dart Programming Language Documentation – https://dart.dev/guides
4. Cloud Firestore Documentation – https://firebase.google.com/docs/firestore
5. Firebase Authentication Documentation – https://firebase.google.com/docs/auth
6. FL Chart Package Documentation – https://pub.dev/packages/fl_chart
7. Flutter Local Notifications Package – https://pub.dev/packages/flutter_local_notifications